Population Dynamics & Projections

Forecasting II: Bayesian Methods in Demography

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Practical Examples – R Lab

- Setup and getting started (HMD and RStan)
- Lee–Carter extension

Figure: GitHub lecture materials

Practical Code and Data (5gb)

Figure: https://shorturl.at/Kg20j

The UN and Probabilistic Projections

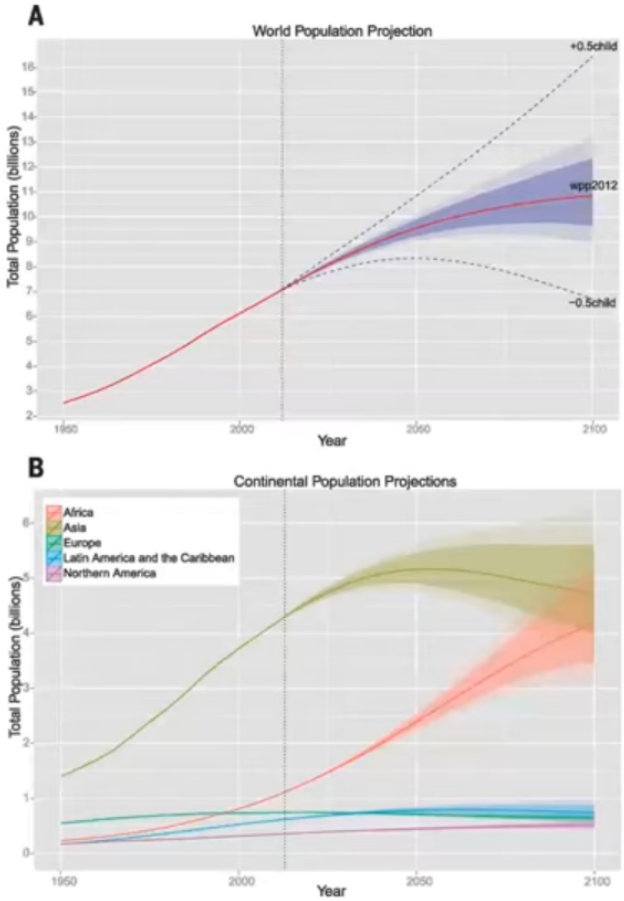

Figure: World Population Projection with probabilistic intervals

- In 2014, UN published probabilistic projections for the first time
- Led by Patrick Gerland at UNPD and Adrian Raftery at UW
- Based on Bayesian hierarchical models for fertility (Alkema et al 2011) and life expectancy (Raftery et al 2013)
- Hugely influential in changing how demographers think about uncertainty in projections

Practical examples - R Lab

This lab demonstrates how to generate probabilistic subnational projections for:

- Total Fertility Rate (TFR) using `bayesTFR`
- Life Expectancy at birth (e_0) using `bayesLife`
- Net Migration Rates using `bayesMig`

The methods follow:

- Ševčíková et al. (2018) for TFR
- Ševčíková and Raftery (2021) for life expectancy
- Azose and Raftery (2015, 2016) for migration

Data: County-level TFR, e_0, and migration rates (1990–2023); national projections from WPP2024.

Motivation

- Approach: generate probabilistic projections of the individual components of the CCM:
  - total fertility rate (TFR),
  - life expectancy at birth (e_0),
  - migration

  $\rightarrow$ probabilistic population projection.

Probabilistic TFR

- National model:
  - Use BHM by Alkema et al. (2011) that
    - estimates different phases of the fertility transition, while sharing information between countries, and
    - generates a set of future TFR trajectories for each country.
- Subnational model:
  - Use Scale AR(1) by Sevčíková et al. (2018) to scale national TFR trajectories to subnational TFR trajectories.
  - Inherits uncertainty from the national projection, while adding slightly more uncertainty contributed by the Scale AR(1).
- Implemented in `bayesTFR` R package.

Probabilistic Mortality

- National model:
  - Generate a set of future female e_0 trajectories for each country via a BHM by Raftery et al. (2013).
  - Generate a set of future male e_0 trajectories via the gap model by Raftery et al. (2014).
  - This results in a joint distribution between the future female and male e_0.
- Subnational model:
  - Use Shift AR(1) by Ševčiková & Raftery (2021) to shift national female e_0 trajectories to subnational e_0 trajectories.
  - Apply the gap model to derive subnational male e_0 trajectories.
  - This results in a joint distribution between the future female and male e_0 for each subnational unit.
- Implemented in `bayesLife` R package.
- The `MortCast` package can be used to disaggregate e_0 into age-specific mortality rates.

Probabilistic Projection of Net Migration

- Model of Azose & Raftery (2015) for projecting international net migration rates.
- Bayesian hierarchical AR(1) model:

$$(r_{c,t} - M_c) = \phi_c(r_{c,t-1} - M_c) + \varepsilon_c$$

- Yields trajectories of future net migration rates for each country.
- Migration rates are converted to net migration counts within CCM.
- Optional constraints on population density and/or zero sum.
- Works well for projecting subnational migration.
- In the BHM setting, migration experience shared among regions.
- Results in a set of future trajectories of net migration rates (converted to counts) for each region.
- Implemented in the `bayesMig` R package.

Probabilistic Population Projection

- Demographic balancing equation:

$$\text{Pop}_t = \text{Pop}_{t-1} + \text{Births}_t - \text{Deaths}_t + \text{Net Migration}_t$$

- Cohort Component Method (CCM), while disaggregating the components into sex- and age-specific counts and rates.
- Probabilistic approach:
  1. Using probabilistic models, generate sets of trajectories for TFR, e_0 and net migration.
  2. Convert each trajectory to age-specific rates and counts.
  3. Apply the CCM to each trajectory of age-specific fertility and mortality rates and migration counts.
  4. Result: Future trajectories of age- and sex-specific population quantities.
- Implemented in the `bayesPop` R package.

Subnational Probabilistic Population Projection

- Methodology applied to 39 counties of Washington state (12 counties smaller than 25,000).
- For each county, TFR and e_0 projections generated to 2050.
- Special treatment of TFR in counties with large college population $\rightarrow$ projected noncollege TFR.
- Net migration BHM excluded 12 small counties from influencing the global experience.
- Cohort-component method applied to each trajectory of the three components of population.